

KINETICS & REACTOR DESIGN — SHORT NOTES

CHE F311 û I-Semester 2026-27 û formulas as taught in the uploaded decks (Slides/)

Companion documents: [KRD-Problems-and-Solutions.pdf](#) (every numerical in the decks, solved) and [krd-answers-check.py](#).

1 û Mole balances, rates, conversion (Ch1-krd (1).pdf)

- Conversion $X = \frac{\text{moles A reacted}}{\text{moles A fed}}$; extent α : $N_i = N_i^0 + \nu_i \alpha$.
- Rate bases: $-r_A$ (mol/vol.t); per mass $-r'_A$, per surface r'' ; $-r'_A W = r'' S = r V$.
- General mole balance: $F_{j0} - F_j + G_j = \frac{dN_j}{dt}$; design equations:

$$\begin{array}{lll} \text{Batch} & \frac{dN_A}{dt} = r_A V & t = N_{A0} \int_0^X \frac{dX}{-r_A V} \\ \text{CSTR} & V = \frac{F_{A0} X}{(-r_A)_{exit}} & \text{PFR} \quad \frac{dF_A}{dV} = r_A, \quad V = F_{A0} \int_0^X \frac{dX}{-r_A} \\ \text{PBR} & \frac{dF_A}{dW} = r'_A & W = F_{A0} \int_0^X \frac{dX}{-r'_A} \end{array}$$

- Space time $\tau = V/v_0$; space velocity $= v_0/V$.

2 û Sizing with conversion (Ch2-krd (1).pdf)

- Levenspiel plot: $F_{A0}/(-r_A)$ vs X ; CSTR = rectangle, PFR = area under curve.
- Numerical integrals: Simpson $\frac{h}{3}(f_0 + 4f_1 + f_2)$; five-point $\frac{h}{3}(f_0 + 4f_1 + 2f_2 + 4f_3 + f_4)$; 3/8-rule $\frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3)$.
- Reactors in series: $F_{Ai} = F_{A0}(1 - X_i)$; CSTR i : $V_i = \frac{F_{A0}(X_i - X_{i-1})}{(-r_A)_i}$.
- PFRs in parallel: efficient split keeps V/F (or τ) equal in each branch.
- Equal-size CSTRs, 1st order: $X = 1 - \frac{1}{(1 + k\tau)^N}$; 2nd order: $\frac{C_{A0}}{C_{AN}} = \left[\frac{-1 + \sqrt{1 + 4kC_{A0}\tau}}{2} \right]^{-N}$ type recursion.
- Best arrangement: $1/(-r_A)$ vs C_A curve; $n > 1$: plug, small mixed, large mixed; $n < 1$ reversed; rectangle-maximisation $-dy/dx = y/x$ for two CSTRs.

3 û Rate laws & stoichiometric tables (Ch3-krd.pdf)

- Relative rates: $\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_P}{d}$.
- Power law $-r_A = kC_A^\alpha C_B^\beta$, overall $n = \alpha + \beta$; Arrhenius $k = Ae^{-E/RT}$, $\ln \frac{k_2}{k_1} = \frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$.
- Reversible: at equilibrium $-r_A \rightarrow 0$ and $K_C = \frac{k_f}{k_r}$ (elementary).
- Batch table ($aA + bB \rightarrow cC + dD$): $N_A = N_{A0}(1 - X)$, $N_B = N_{A0}(\Theta_B - \frac{b}{a}X)$, $N_C = N_{A0}(\Theta_C + \frac{c}{a}X)$; liquid: $C_i = N_i/V_0$.
- Flow table: same with F 's; gas phase $C_i = C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \frac{T_0}{T} \frac{P}{P_0}$, $\varepsilon = y_{A0}\delta$, $\delta = \frac{c + d - b - a}{a}$.
- Batch gas, const P, T : $V = V_0(1 + \varepsilon X)$, $C_A = \frac{N_{A0}(1 - X)}{V_0(1 + \varepsilon X)}$.

4 û Isothermal design (Ch4-_Krd.pdf)

- Algorithm: mole balance $\rightarrow$ rate law $\rightarrow$ stoichiometry $\rightarrow$ combine $\rightarrow$ evaluate.
- Batch 1st order $t = \frac{1}{k} \ln \frac{1}{1 - X}$; 2nd order $t = \frac{1}{kC_{A0}} \frac{X}{1 - X}$.
- CSTR 1st order $X = \frac{k\tau}{1 + k\tau}$; 2nd order $X = \frac{(1 + 2Da) - \sqrt{1 + 4Da}}{2Da}$, $Da = k\tau C_{A0}$.
- Damköhler: $Da_1 = k\tau$; $Da \leq 0.1 \Rightarrow X < 0.1$; $Da \geq 10 \Rightarrow X > 0.9$.
- CSTRs in series, 1st order: $X = 1 - \frac{1}{(1 + k\tau_i)^N}$; parallel CSTRs behave as one big CSTR.
- PFR gas, 2nd order, ε : $V = \frac{v_0}{kC_{A0}} \left[2\varepsilon(1 + \varepsilon) \ln(1 - X) + \varepsilon^2 X + \frac{(1 + \varepsilon)^2 X}{1 - X} \right]$.
- Pressure drop (Ergun $\rightarrow$): $\frac{dp}{dW} = -\frac{\alpha}{2p}$, $p = \frac{P}{P_0}$; $\varepsilon = 0$: $p = (1 - \alpha W)^{1/2}$; coupled with $\frac{dX}{dW} = f(X, p)$; pipe form $\frac{dP}{dz} = -\frac{\beta_0 P_0}{P} (1 + \varepsilon X)$, $\beta_0 = \frac{G(1 - \phi)}{\rho_0 D_p \phi^3} \left[\frac{150(1 - \phi)\mu}{D_p} + 1.75G \right]$.
- $C_A = C_{A0} \frac{1 - X}{1 + \varepsilon X}$ p (gas, const T).

5 û Rate-data analysis (Ch5_- krd.pdf)

- Differential: plot $\ln(-dC_A/dt) = \ln k + \alpha \ln C_A$; finite differences: initial $\frac{-3C_0+4C_1-C_2}{2\Delta t}$, interior $\frac{C_{i-1}-C_{i+1}}{2\Delta t}$, final $\frac{C_{N-2}-4C_{N-1}+3C_N}{2\Delta t}$.
- Integral: guess order, linearise: 0th C_A vs t ; 1st $\ln C_A$; 2nd $1/C_A$; $A+B$, $M = C_{B0}/C_{A0} \neq 1$: $\ln \frac{C_A}{C_B} = \ln M + C_{A0}(M-1)kt$.
- Half-life: $t_{1/2} = \frac{2^{n-1}-1}{(n-1)k} C_{A0}^{1-n}$; $\ln t_{1/2} = (1-n) \ln C_{A0} + \text{const}$; 1st order $t_{1/2} = \ln 2/k$.
- Initial rates, differential reactor: $-r'_A = \frac{F_{A0}-F_A}{W} \approx \frac{F_P}{W}$.
- Nonlinear regression: minimise $s^2 = \sum (C_{m,i} - C_{e,i})^2$; Michaelis-Menten $-r_S = \frac{k_3 C_S C_E}{C_S + C_M}$, integrated $k_3 C_E t = (C_{S0} - C_S) + C_M \ln \frac{C_{S0}}{C_S}$.
- Variable volume (batch, const P): $X = \frac{(V/V_0)-1}{\varepsilon_A}$.

6 û Multiple reactions (Ch6_krd.pdf)

- Instantaneous selectivity $S_{D/U} = r_D/r_U$; overall $\tilde{S}_{D/U} = F_D/F_U$ (batch: N_D/N_U).
- Yields: $\tilde{Y}_D = r_D/(-r_A)$ instantaneous; overall $F_D/(F_{A0} - F_A)$.
- Parallel, one reactant: $a = \alpha_1 - \alpha_2 > 0 \Rightarrow$ keep C_A high (PFR/batch, high P , no inerts); $a < 0 \Rightarrow$ keep low (CSTR, inerts, dilution).
- Temperatures: $E_D > E_U \Rightarrow$ run hot; $E_U > E_D \Rightarrow$ run cool (but reactive).
- Two reactants: signs of $(\alpha_1 - \alpha_2)$, $(\beta_1 - \beta_2)$ choose contacting pattern (semibatch, side streams, small CSTRs in series).
- Series $A \xrightarrow{k_1} B \xrightarrow{k_2} C$: maximise B by space/batch time; C_B^{max} at $t = \frac{\ln(k_2/k_1)}{k_2 - k_1}$.
- Net rates: $r_j = \sum_i \nu_{ji} r_i$; product distribution $\varphi(R/A) = \frac{dC_R}{-dC_A}$; PFR $C_R = \int \varphi d(-C_A)$ -area, MFR rectangle, staged = steps.

7 û Non-isothermal design (Ch7_krd.pdf)

- Energy balance (steady, no shaft): $\dot{Q} - \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = 0$; $H_i(T) = H_i^\circ(T_R) + C_{p,i}(T - T_R)$.
- Adiabatic: $X = \frac{\sum \Theta_i C_{p,i}(T - T_0)}{-[\Delta H_{Rx}^\circ(T_R) + \Delta C_p(T - T_R)]}$; often $T = T_0 + (\text{slope})X$.
- $K_C(T) = K_C(T_1) \exp \left[\frac{\Delta H_{Rx}}{R} \left(\frac{1}{T_1} - \frac{1}{T} \right) \right]$; $X_e = \frac{K_C}{1+K_C}$ for $A \rightleftharpoons B$.
- Tubular with exchange: $\frac{dT}{dV} = \frac{Ua(T_a - T) + (-r_A)(-\Delta H_{Rx})}{\sum F_i C_{p,i}}$; coolant co-current $\frac{dT_a}{dV} = \frac{Ua(T - T_a)}{\dot{m}_c C_{p,c}}$.
- CSTR with cooling: solve $X = \frac{k\tau}{1+k\tau}$ with $X = \frac{\sum \Theta_i C_{p,i}(T - T_0) + \kappa(T - T_a)}{F_{A0}(-\Delta H_{Rx})}$, $\kappa = UA/F_{A0}$ -style term.
- Multiple steady states: $G(T) = \frac{-\Delta H_{Rx}}{1+1/(k\tau)}$ -type heat-generated vs $R(T) = C_{p0}(1 + \kappa)(T - T_c)$; ignition/extinction; runaway when tangency lost; stability via $S' = \frac{E}{RT_s^2} G(T_s) - C_{p0}(1 + \kappa)$.
- Interstage cooling/heating between adiabatic reactors; optimum feed temperature maximises X for fixed V .

8 û Catalysis (Catalysis_Krd.pdf)

- Steps: external diffusion $\rightarrow$ internal diffusion $\rightarrow$ adsorption $\rightarrow$ surface reaction $\rightarrow$ desorption; rate-limiting-step algorithm with site balance $C_t = C_v + \sum C_{i,s}$.
- Langmuir $C_{A,s} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B}$; L-H vs Eley-Rideal forms (e.g. $-r'_T = \frac{k P_T P_{H_2}}{1 + K_B P_B + K_T P_T}$).
- Thiele modulus (1st order, sphere) $\phi_s = R\sqrt{k_1/D_e}$; $\eta = \frac{3}{\phi^2}(\phi \coth \phi - 1)$; slab $\eta = \tanh \phi / \phi$.
- Profile: $\frac{C_A}{C_{As}} = \frac{R}{r} \frac{\sinh(\phi r/R)}{\sinh \phi}$.
- Weisz-Prater $C_{WP} = \eta \phi^2 = \frac{-r'_A(\text{obs}) R^2}{D_e C_{As}}$; $C_{WP} \ll 1$ no internal limitation, $\gg 1$ limited.
- Overall $\Omega = \frac{\eta}{1 + \eta \phi^2 S_a \rho_b / k_c}$; Mears: $\frac{k C_{Ab}^{n-1} R^2 \rho_b S_a E}{b D_e T^2} < 0.15$ to neglect external gradients (ratio $r_{obs}/(k_c C_{Ab}) < 0.15$ style).
- Packed bed: workhorse; adiabatic/non-adiabatic, reverse-flow, counter-flow configurations.

9 û Non-ideal reactors / RTD (Non_ideal_KRD.pdf)

- $E(t) = \frac{C(t)}{\int_0^\infty C dt}$; $F(t) = \int_0^t E dt$; $t_m = \int_0^\infty t E dt = V/v_0$; $\sigma^2 = \int (t - t_m)^2 E dt$.
- Ideal: PFR $E = \delta(t - \tau)$; CSTR $E = e^{-t/\tau}/\tau$.
- Segregation: $\bar{X} = \int_0^\infty X_{batch}(t) E(t) dt$; $n > 1$: $\bar{X} > X_{ideal-mixed}$; $0 < n < 1$ opposite.
- Tanks-in-series: $E(t) = \frac{t^{n-1}}{(n-1)! \tau_i^n} e^{-t/\tau_i}$, $n = \tau_m^2/\sigma^2$; n -tank conversion for 1st order $X = 1 - (1 + k\tau_i)^{-n}$.
- Dispersion (closed vessel, 1st order): $X = 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Peq/2} - (1-q)^2 e^{-Peq/2}}$, $q = \sqrt{1 + 4Da/Pe}$, $Da = k\tau$, $Pe = UL/D_a$;

$$\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2}(1 - e^{-Pe}); \text{ link } n = Pe/2 + 1.$$

- Two-parameter models: real CSTR = ideal volume $V_s = \alpha V$ + dead zone + bypass $\beta = v_b/v_0$; tracer response $C_T/C_{T0} = 1 - (1 - \beta)e^{-(1-\beta)t/(\alpha\tau)}$ -type; two-CSTR interchange parameters (α, β) from $m_{1,2}$.

Compiled from the uploaded decks; the questions document (**KRD-Problems-and-Solutions.pdf**) applies every one of these formulas to the numerals printed in the slides, and **krd-answers-check.py** reproduces all numbers.